

1. Title and introduction Project title, objective, and business questions.
2. Dataset description UCI Online Retail dataset, transaction records, and important columns.
3. Data cleaning and methodology Missing values, cancelled orders, data types, and revenue calculation. Describe only the steps you actually performed.
4. Exploratory data analysis Your five charts, with a short explanation below each chart.
5. Key findings and business recommendations Sales performance, top products, customer behavior, and RFM segmentation.
6. Conclusion and limitations Summarize what you learned and mention that findings depend on the data-cleaning rules used.

E-Commerce Sales & Customer Behavior Analysis

Exploratory Data Analysis (EDA) Report

Project Objective

The objective of this project is to analyze e-commerce transaction data to understand sales performance, product demand, customer purchasing behavior, and customer segments. The analysis uses Python to identify patterns and develop data-driven business recommendations.

Dataset: UCI Online Retail Dataset

Tools: Python, Pandas, Matplotlib, Google Colab

Analysis Areas: Data Cleaning, Exploratory Data Analysis, Sales Analysis, Customer Analysis, and RFM Segmentation

Author: N.Endhuja

Date: October 2026

1. Introduction

E-commerce businesses generate large amounts of transaction data. Analyzing this data helps businesses understand sales performance, product demand, and customer purchasing behavior.

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Online Retail dataset using Python. The analysis focuses on identifying sales trends, top-performing products and countries, customer contributions, and customer segments.

Project Objectives

- Analyze overall sales revenue and monthly sales trends.
- Identify popular products and important markets.
- Examine customer purchasing behavior.
- Segment customers using RFM analysis.
- Develop data-driven business recommendations.

2. Dataset Description

This project uses the Online Retail dataset published by the UCI Machine Learning Repository.

Dataset source: <https://archive.ics.uci.edu/dataset/352/online+retail>

The dataset contains transactions from a UK-based online retail business.

Important Columns

- **InvoiceNo:** Invoice identifier for each transaction.
- **StockCode:** Product identifier.
- **Description:** Product name or description.
- **Quantity:** Number of units purchased.

- **InvoiceDate:** Date and time of the transaction.
- **UnitPrice:** Price per unit.
- **CustomerID:** Customer identifier.
- **Country:** Customer's country.

Revenue Calculation

Revenue was calculated using:

Revenue = Quantity × UnitPrice

The reported results depend on the data-cleaning and filtering rules used in the analysis.

3. Key Findings

The analysis produced the following results:

- **Total Revenue:** £10,666,684.54
- **Unique Customers:** 4,338
- **Highest-Revenue Month:** November 2011
- **November 2011 Revenue:** £1,509,496.33
- **Top Country by Revenue:** United Kingdom
- **Best-Selling Product by Quantity:** PAPER CRAFT , LITTLE BIRDIE

These findings summarize the sales performance and customer base observed in the analyzed dataset.

4. Business Recommendations

1. **Inventory planning:** Monitor popular products and maintain suitable stock levels.
2. **Seasonal planning:** Analyze high-revenue periods to support future promotions and inventory decisions.
3. **Customer retention:** Use customer purchasing patterns to design relevant loyalty campaigns.
4. **Customer reactivation:** Consider targeted campaigns for customers who have not purchased recently.
5. **Performance monitoring:** Track monthly revenue, product demand, and customer activity regularly.

These are potential actions suggested by the analysis and should be tested against future business results.

5. Conclusion

This project demonstrates how Python can be used to clean, analyze, and visualize e-commerce transaction data. The analysis explored sales trends, product performance, country-level revenue, customer contributions, and RFM customer segments.

The findings can help inform inventory planning, marketing activities, and customer-retention strategies. The results should be interpreted in the context of the dataset and the cleaning rules applied.

6. Tools Used

- Python
- Pandas
- Matplotlib
- Google Colab
- Jupyter Notebook

```
from IPython.display import Image, display
display(Image(filename="01_monthly_sales_trend.png", width=700))
```

[Show hidden output](#)

```
from IPython.display import Image, display
display(Image(filename="02_top_10_products.png", width=700))
```

[Show hidden output](#)

```
from IPython.display import Image, display  
  
display(Image(filename="03_top_10_countries.png", width=700))
```

[Show hidden output](#)

```
from IPython.display import Image, display  
  
display(Image(filename="04_top_10_customers.png", width=700))
```

[Show hidden output](#)

```
from IPython.display import Image, display  
  
display(Image(filename="05_customer_segments.png", width=700))
```

[Show hidden output](#)